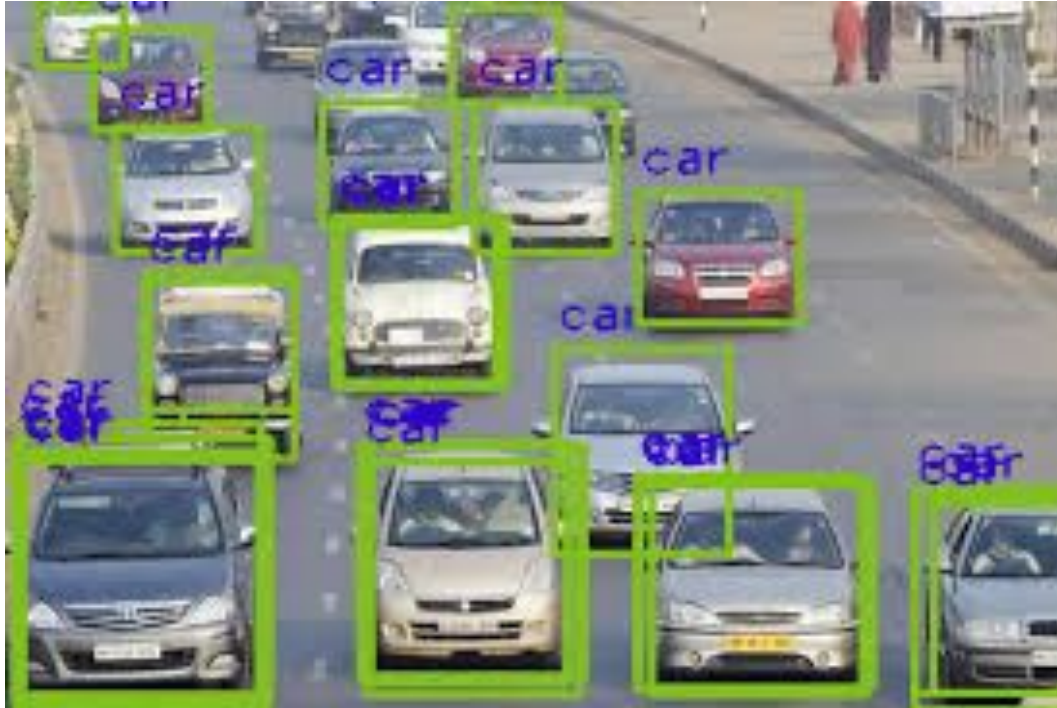


Detection and segmentation

Mark Blumenau

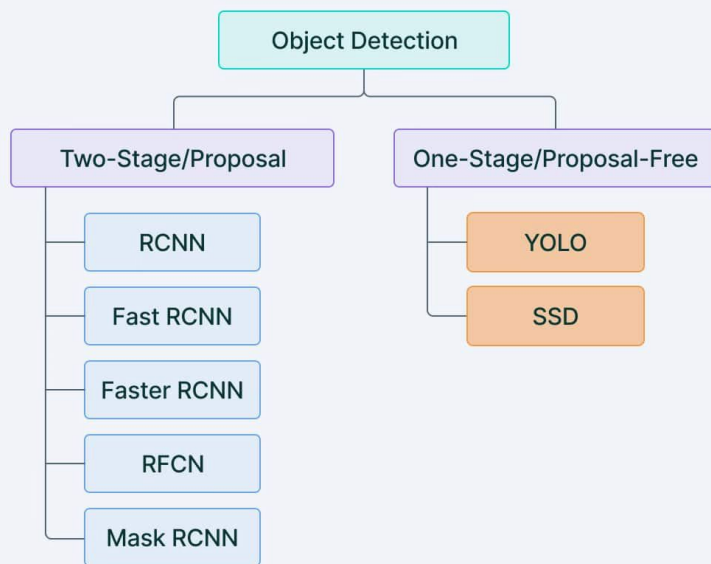
Is classification useless?

What is detection?

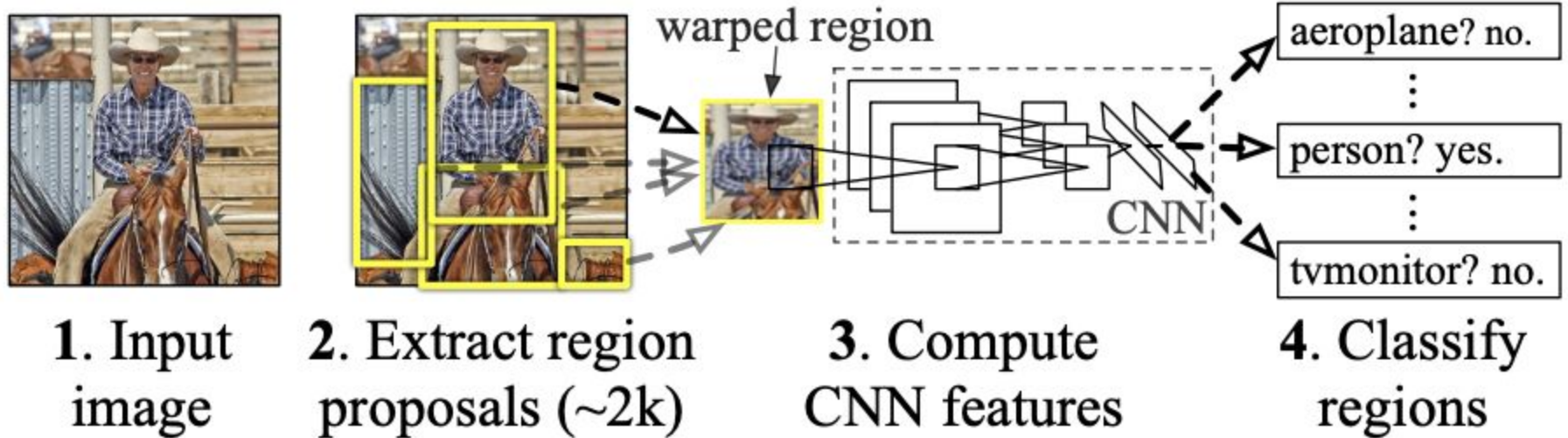


How can we do it?

One and two stage detectors

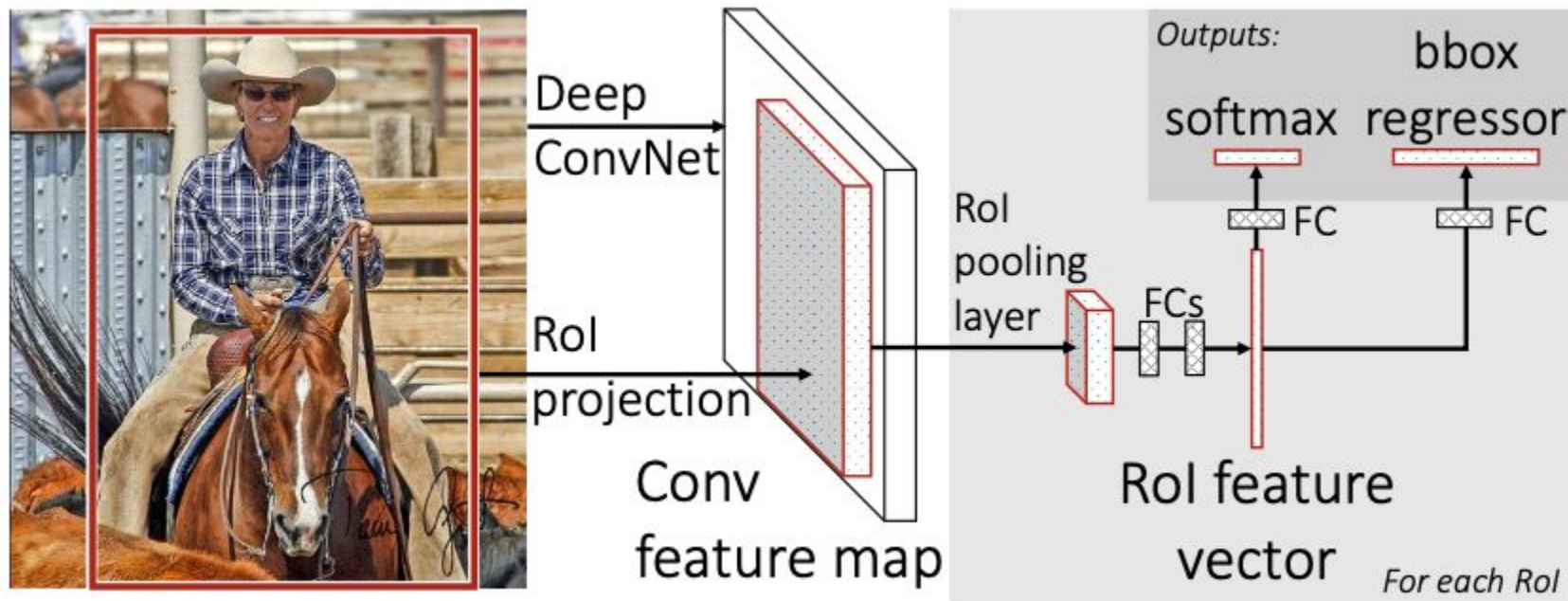


Two shot (stage) detector: R-CNN 2013

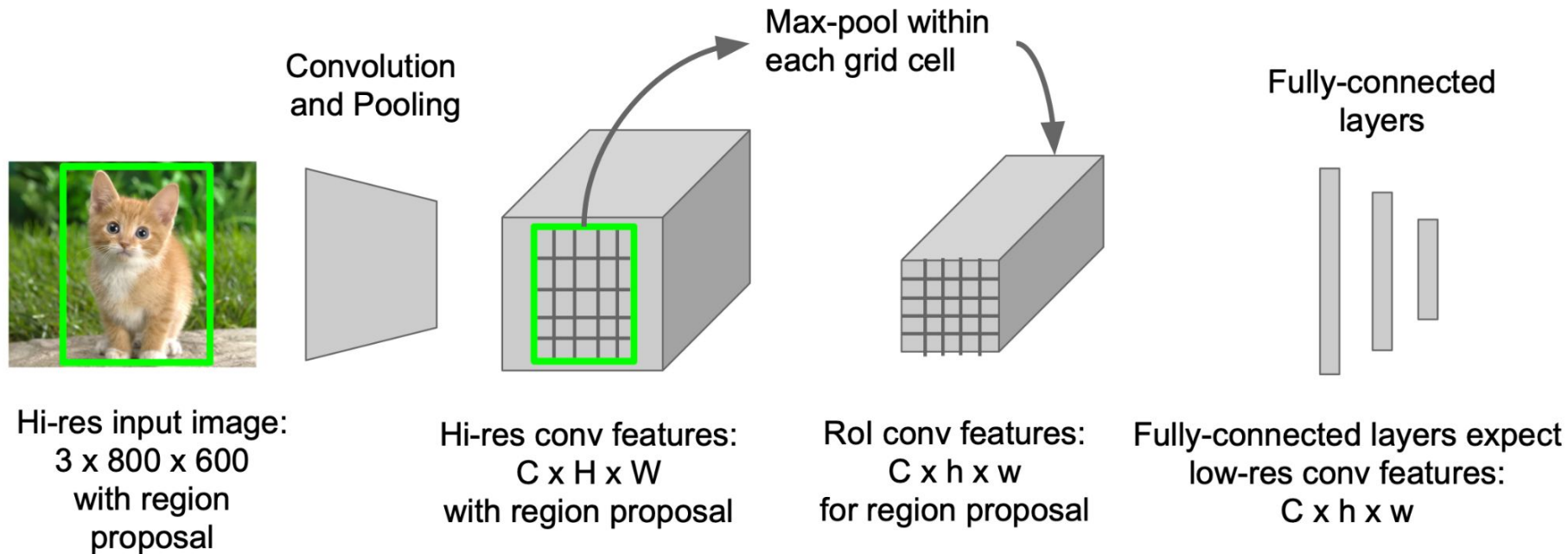


Why is it not the best way to do so?

Fast R-CNN 2015: we need to go faster



ROI Projection



Our new loss function

$$L(p, u, t^u, v) = L_{\text{cls}}(p, u) + \lambda[u \geq 1]L_{\text{loc}}(t^u, v)$$

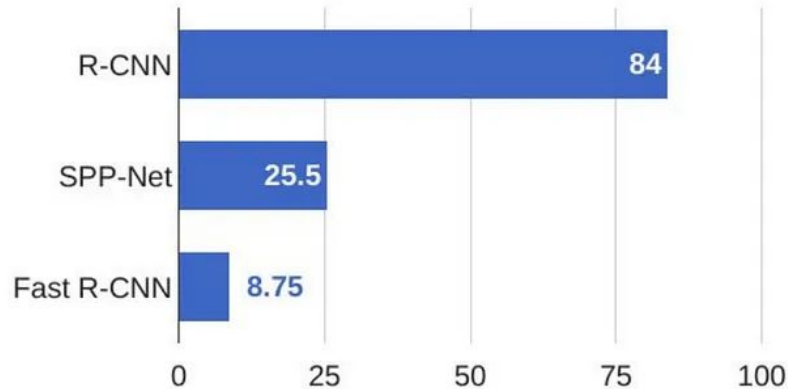
$$L_{\text{loc}}(t^u, v) = \sum_{i \in \{x, y, w, h\}} \text{smooth}_{L_1}(t_i^u - v_i),$$

in which

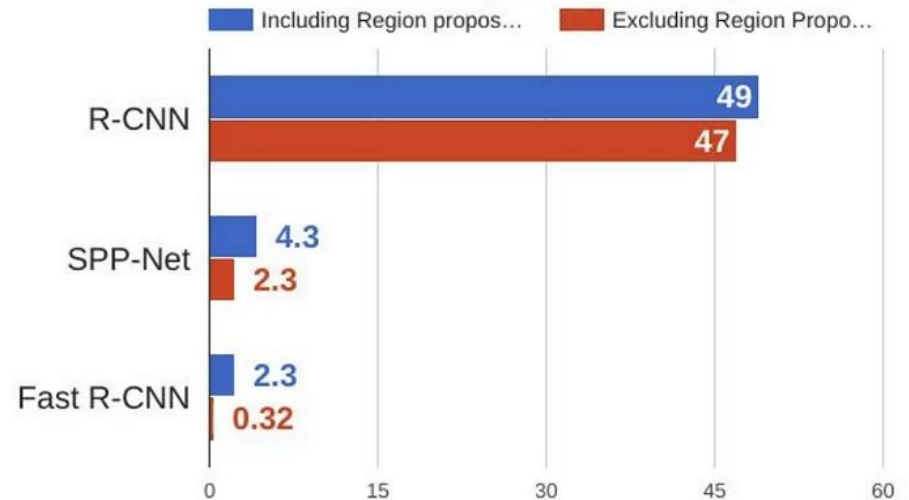
$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise,} \end{cases}$$

Speed comparison

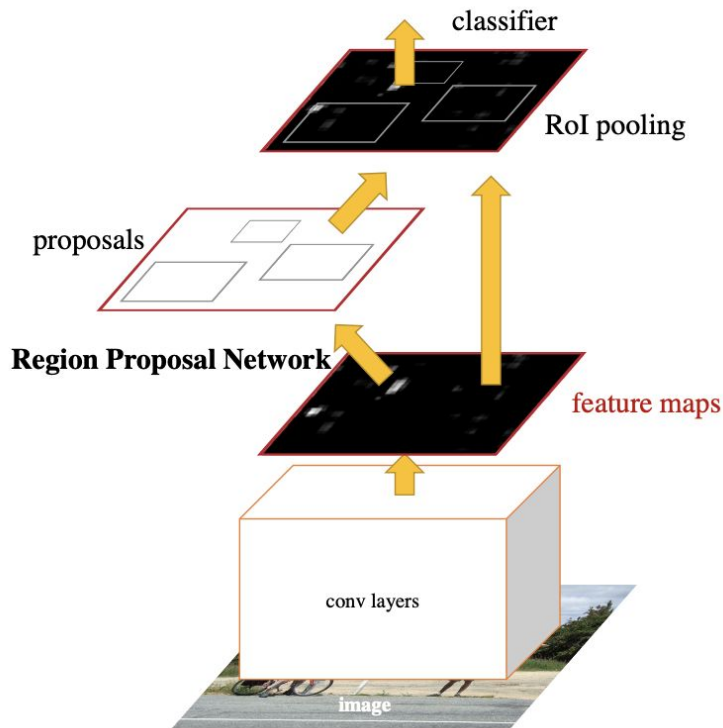
Training time (Hours)



Test time (seconds)



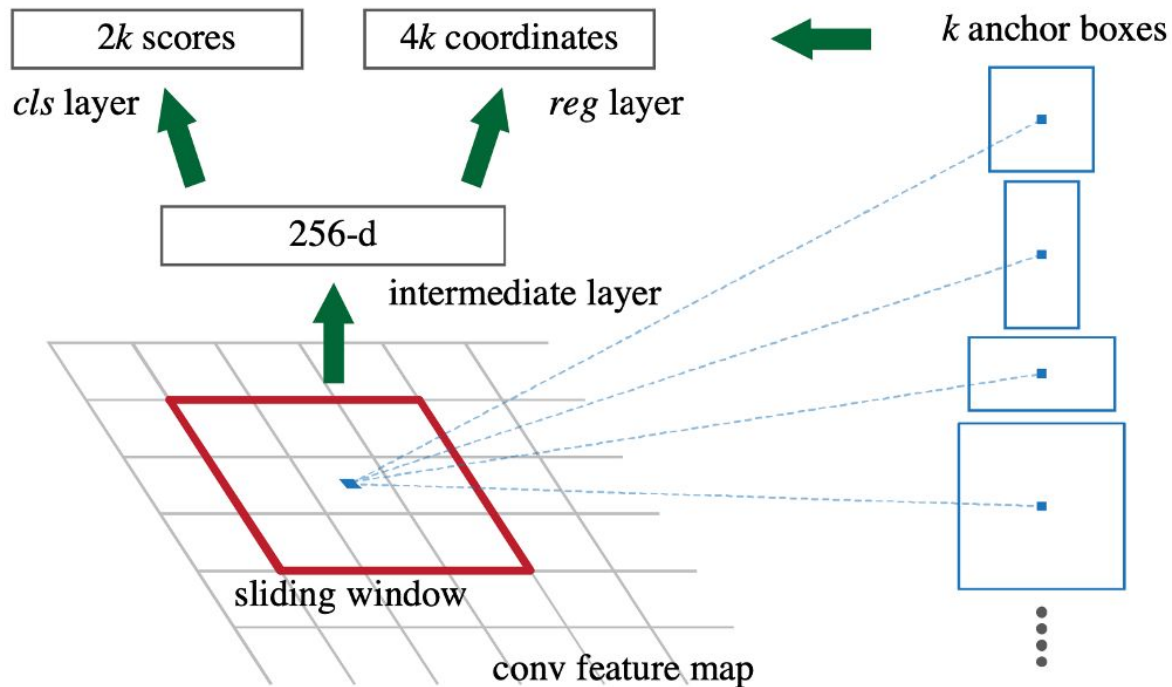
Faster R-CNN 2015 (+2 month): what if we reject classical CV?



Region proposal network

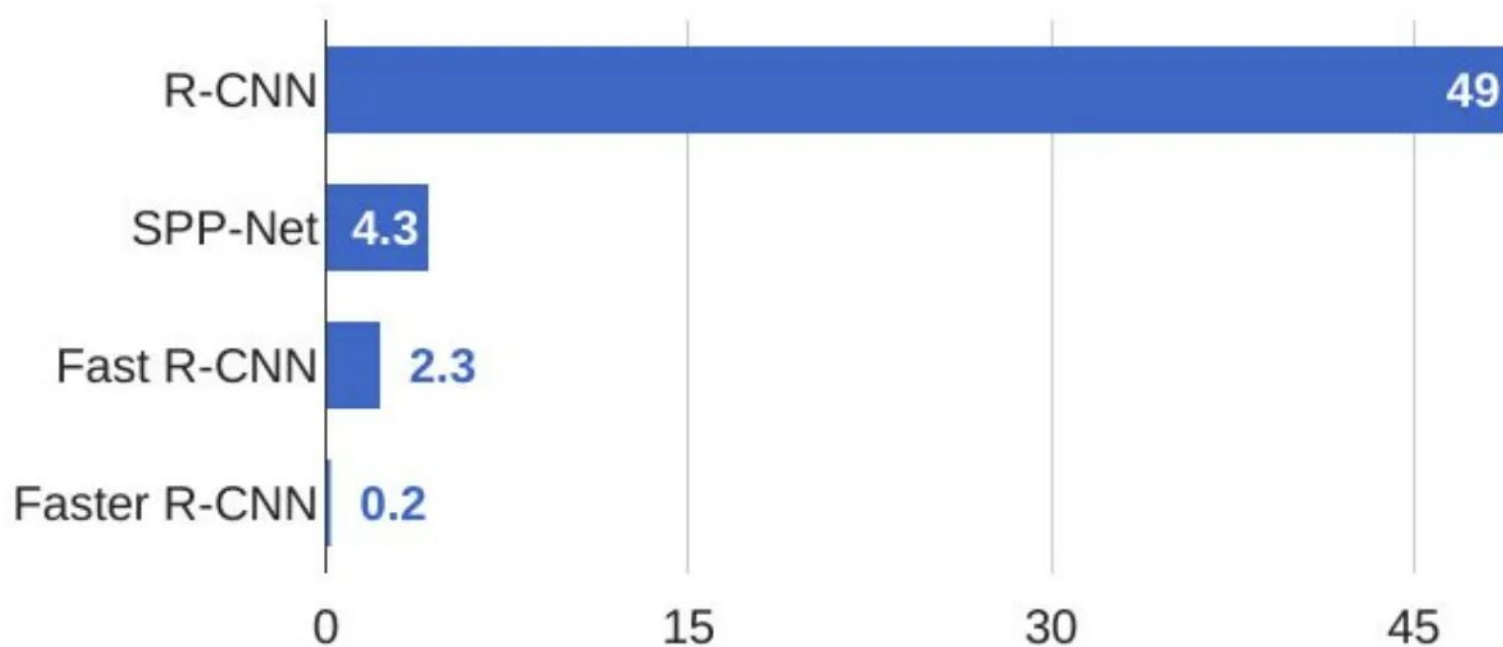
- Take the last tensor from the convolutional network
- Apply a 3x3 convolutional layer with a large number of channels (256 or 512)
- For each point, take 9 rectangles with different side lengths and aspect ratios (anchors)
- For each, predict:
 - Whether there is an object there
 - 4 corrections to the sides

Region proposal network

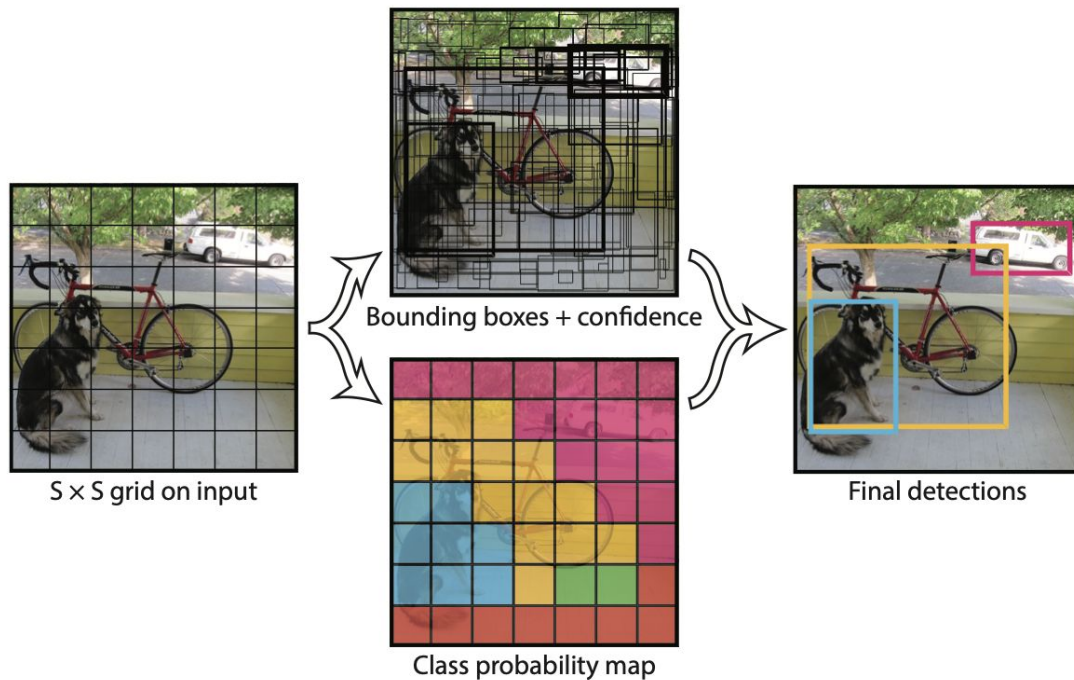


The speed!

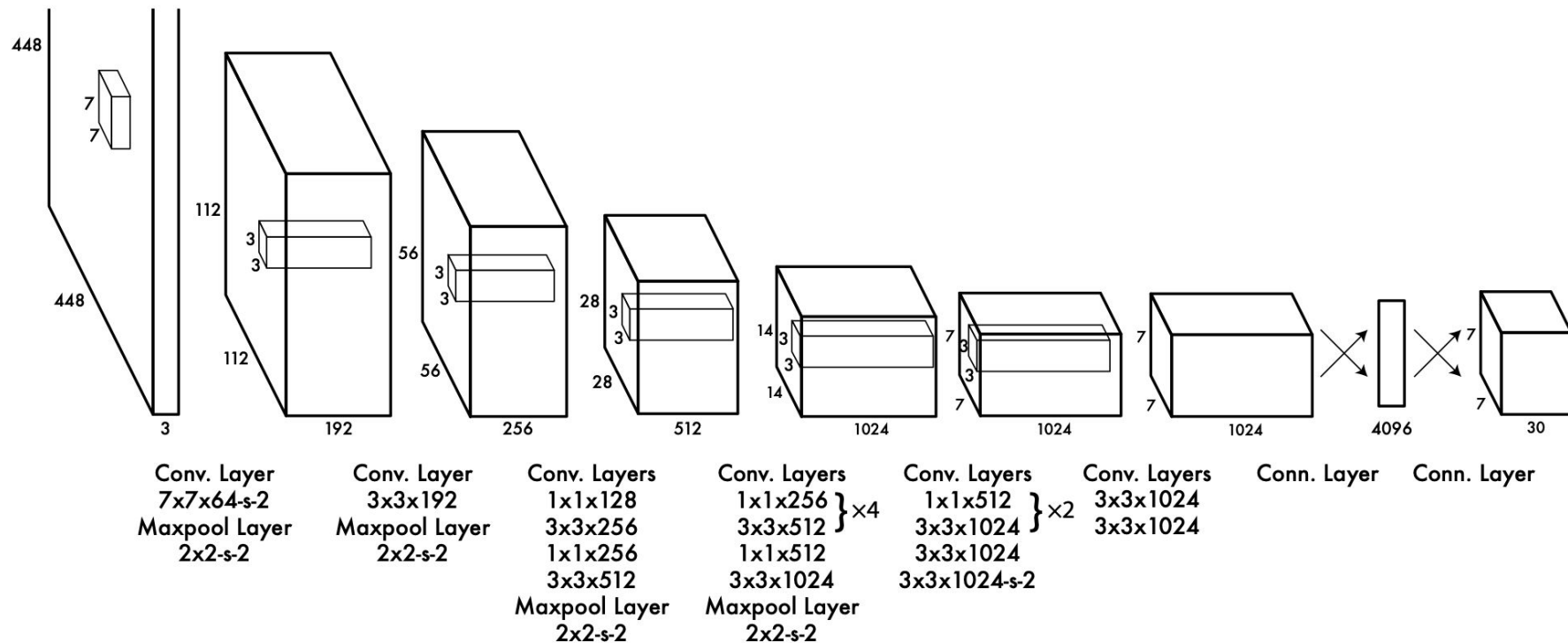
R-CNN Test-Time Speed



But can we get realtime?



How does the arch look like?



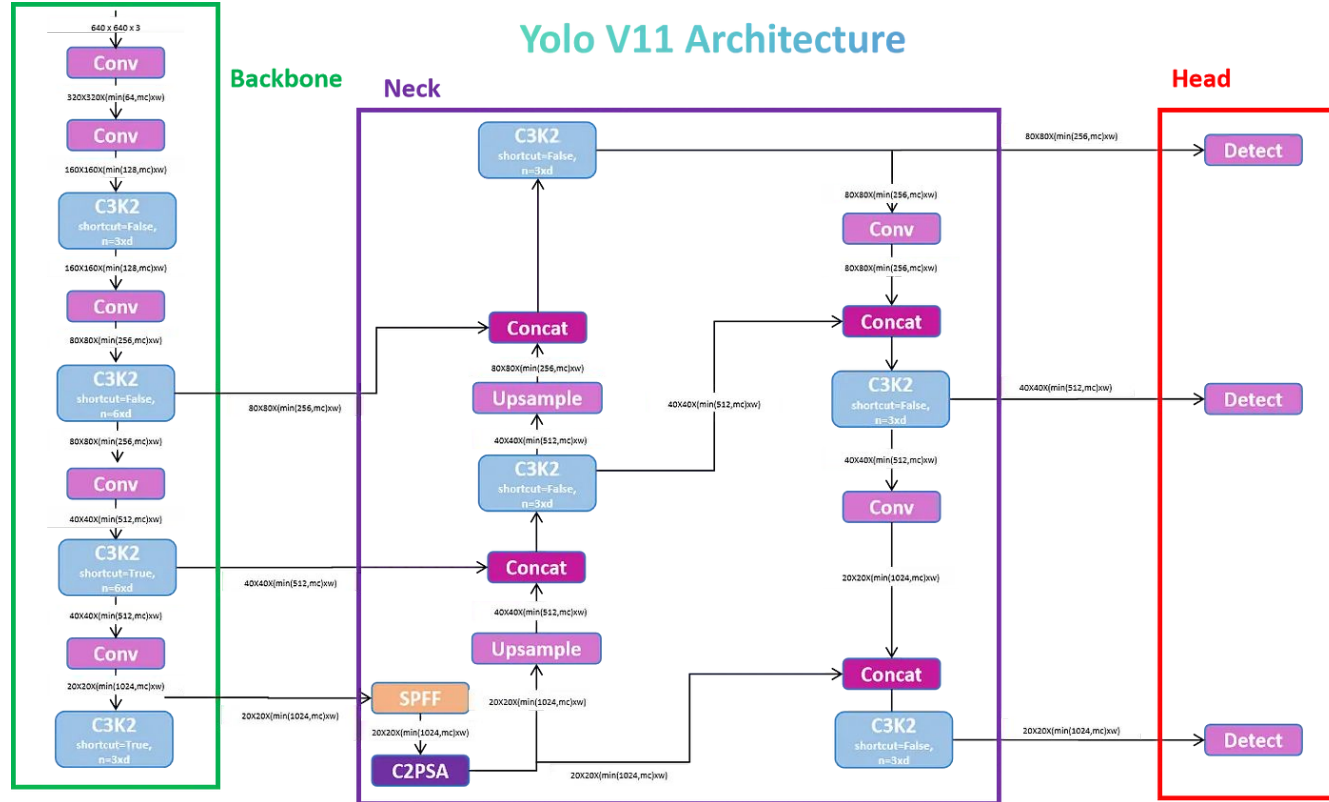
How can we teach it?

$$\begin{aligned} & \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \\ & + \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right] \\ & + \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2 \\ & + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2 \\ & + \sum_{i=0}^{S^2} \mathbb{1}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2 \end{aligned}$$

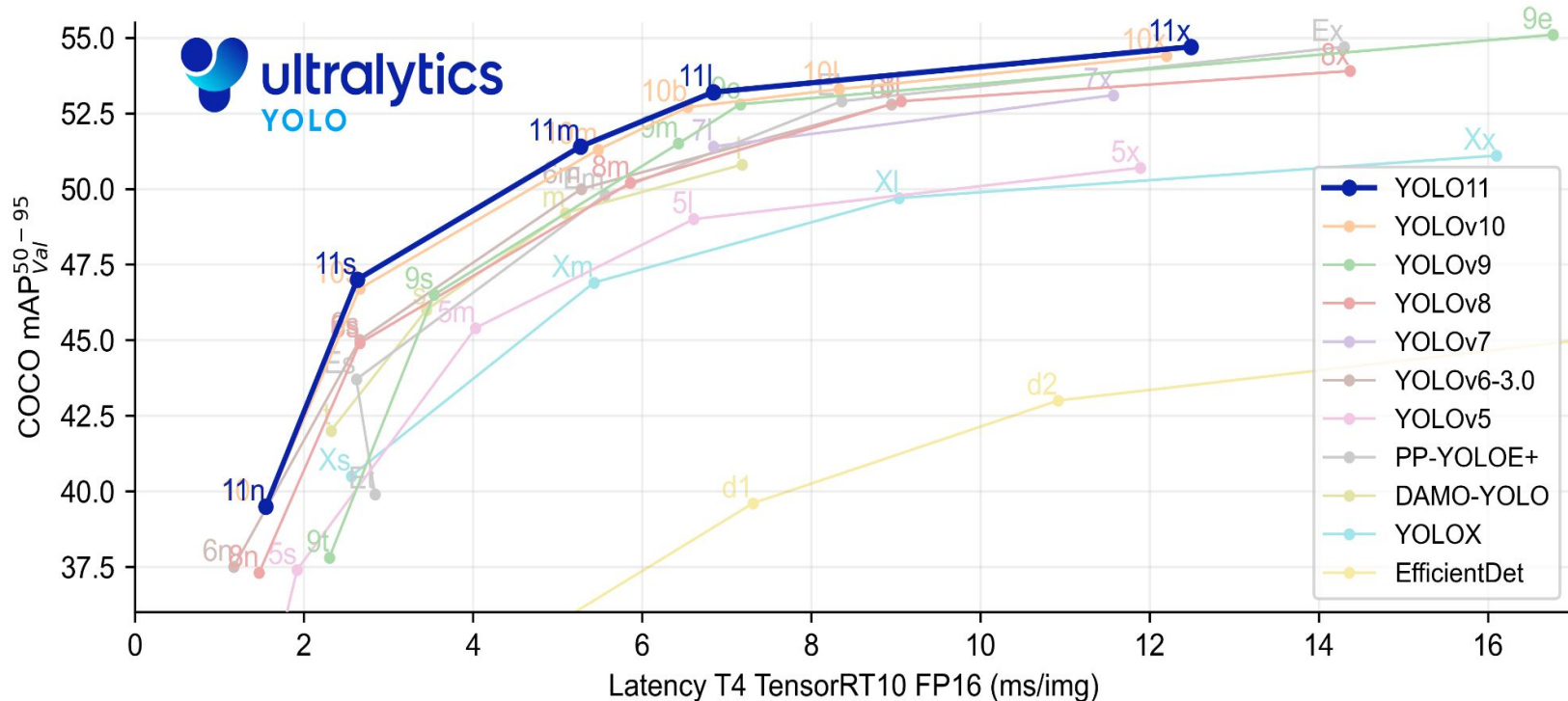
Why did we do all of it?

Real-Time Detectors	Train	mAP	FPS
100Hz DPM [31]	2007	16.0	100
30Hz DPM [31]	2007	26.1	30
Fast YOLO	2007+2012	52.7	155
YOLO	2007+2012	63.4	45
Less Than Real-Time			
Fastest DPM [38]	2007	30.4	15
R-CNN Minus R [20]	2007	53.5	6
Fast R-CNN [14]	2007+2012	70.0	0.5
Faster R-CNN VGG-16[28]	2007+2012	73.2	7
Faster R-CNN ZF [28]	2007+2012	62.1	18
YOLO VGG-16	2007+2012	66.4	21

Yolo 11 aka what we do in 2025



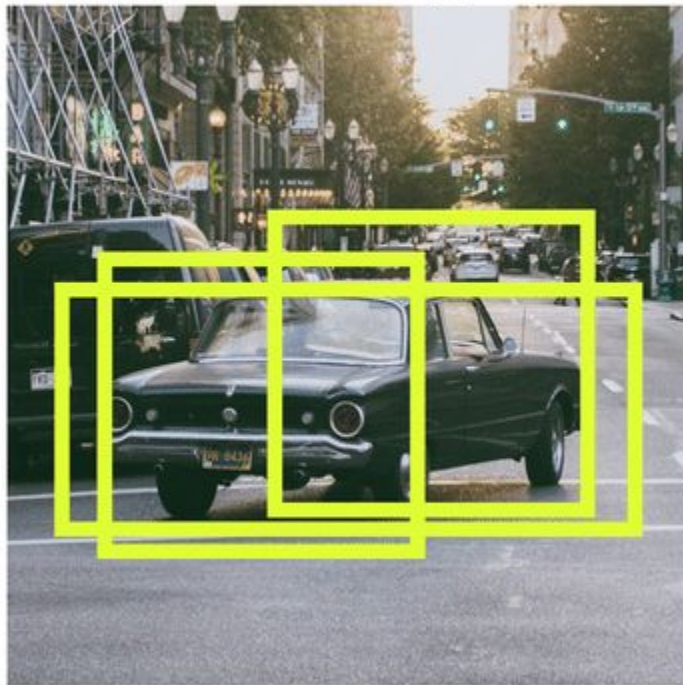
Quality comparison



<https://github.com/ultralytics/ultralytics>

Non-maximum suppression (NMS)

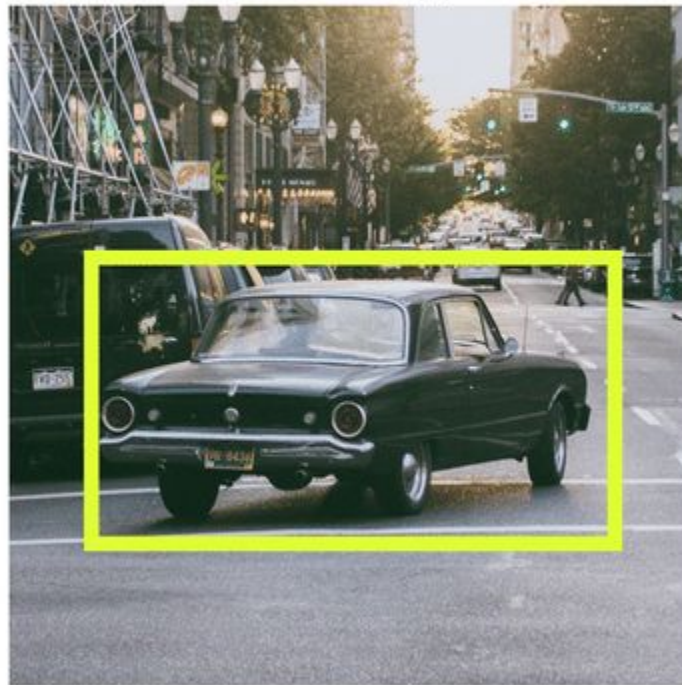
Before non-max suppression



Non-Max
Suppression



After non-max suppression



Non-maximum suppression (NMS)

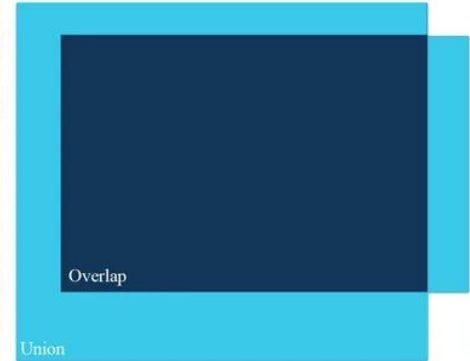
- The model outputs a list of rectangles with confidence scores for class C.
- We go through the list in descending order of confidence.
- For each rectangle, we remove all subsequent ones with an Intersection over Union (IoU) > 0.5 .

Metric (one of)

- 1) We obtain a list of bboxes.
- 2) If $\text{IoU}(y, \text{pred}) > 0.5 \rightarrow$ bbox is correct.
- 3) If the detection class is also correct, we set True Positive (TP).
- 4) If $\text{IoU} < 0.5$ or the box is duplicated – False Positive (FP).
- 5) If $\text{IoU} > 0.5$, but the class is incorrect or there is no bbox at all – False Negative (FN)



$$\text{IoU} = \frac{\text{area of overlap}}{\text{area of union}}$$



PR curve

Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

AP = Average Precision = AUC PR

mAP = mean Average Precision

